

Database Table Structure

Tuition management app — Django tuition app

Source of truth: `tuition/models.py`

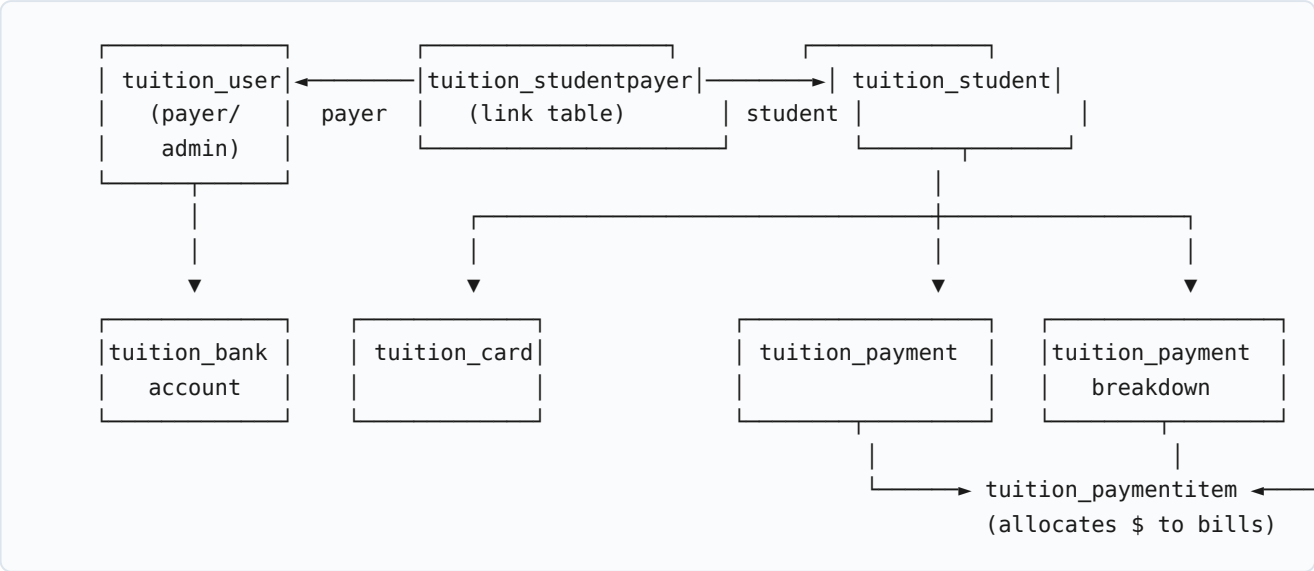
User table: `tuition_user` (custom model, `AUTH_USER_MODEL = 'tuition.User'`)

Default DB (local): SQLite (`db.sqlite3`) · **Staging / production:** PostgreSQL

Table naming: Django default — `tuition_` + lowercase model name

At a glance — main relationships

Students link to payer users through StudentPayer. Payments tie to students (and payers); bill lines use PaymentBreakdown and PaymentItem.



1. Identity & students

tuition_user — Payers and admins

Extends Django `AbstractUser` (login, names, email, staff flags, etc.) plus app-specific fields.

Column	Type / values	Notes
id	BigInt	PK
username, password, email, first_name, last_name	Text	Standard Django auth
is_staff, is_active, is_superuser, date_joined, last_login	Various	Standard Django auth
user_type	payer admin	Role
user_id	Char(30)	Unique
stripe_customer_id	Char(64)	Optional

tuition_student

Column	Type / values	Notes
id	BigInt	PK
student_id	Char(100)	Unique
first_name, last_name	Char(100)	
date_of_birth	Date	
grade	Integer	Grade level
status	active inactive	Default: active
notes	Text	Admin notes
current_balance	Decimal(10,2)	Default 0.00
due_date	Date	Optional
created_at, updated_at	DateTime	Auto

tuition_studentpayer — Student ↔ payer link

Column	Type / values	Notes
id	BigInt	PK
student_id	FK → student	FK CASCADE
payer_id	FK → user	FK CASCADE
relationship	mother father guardian other	
is_primary	Boolean	Default false
created_at, updated_at	DateTime	
Unique together: (student, payer)		

2. Payments & billing

tuition_payment

Column	Type / values	Notes
id	BigInt	PK
student_id	FK → student	FK
payer_id	FK → user	Optional
amount	Decimal(10,2)	
payment_date	DateTime	
status	pending completed failed	
payment_method	Char(50)	e.g. cash, check
notes	Text	Admin notes
bank_account_id	FK → bankaccount	Optional, PROTECT
routing_number, account_number, account_type	Text	Legacy / snapshot
submitted_at	DateTime	Auto on create
receipt_number	Char(50)	Unique
currency	Char(3)	Default USD
exchange_rate, exchange_rate_date	Decimal / DateTime	FX
original_amount, original_currency	Decimal / Char(3)	FX

tuition_paymentbreakdown — Bill line items

Column	Type / values	Notes
id	BigInt	PK
student_id	FK → student	
description	Char(255)	
amount, currency	Decimal / Char(3)	
due_date, date_incurred, late_date	Date	late_date auto if omitted
is_paid	Boolean	
payment_status_override	unpaid partially_paid paid	Manual override
show_in_payment_history	Boolean	
created_at, updated_at	DateTime	

tuition_paymentitem — Payment applied to a bill line

Column	Type / values	Notes
id	BigInt	PK
payment_id	FK → payment	
breakdown_item_id	FK → paymentbreakdown	
amount_paid, currency	Decimal / Char(3)	
created_at	DateTime	
Unique together: (payment, breakdown_item)		

tuition_paymentreceipt

Column	Notes
payment_id	OneToOne → payment
file	Uploaded file path
generated_at	DateTime

tuition_paymentreminder

Column	Notes
payment_id	FK → payment
reminder_date, sent, sent_at, created_at	Reminder tracking

tuition_paymentplan & tuition_paymentinstallment

Table	Main columns
paymentplan	student_id, total_amount, number_of_installments, start_date, end_date, status, timestamps
paymentinstallment	payment_plan_id, amount, due_date, payment_id (OneToOne, optional), status, timestamps

3. Payment methods & vendors

tuition_bankaccount

Column	Notes
user_id	FK → user
nickname, account_type	checking savings
last4, provider_token	Tokenized; not full account number
stripe_payment_method_id, created_at	

tuition_card

Column	Notes
user_id	FK → user
nickname, last4, brand, exp_month, exp_year	
stripe_payment_method_id, created_at	

tuition_vendor

Column	Notes
name, email, bill_vendor_id	External billing integration

4. Onboarding & account security

tuition_accountrequest

Column	Notes
first_name, last_name, child_first_name, child_last_name	Applicant info
email, student_names, submitted_at	

tuition_passwordreset

Column	Notes
user_id, token (unique), created_at, used	Token expires after 24h (app logic)

tuition_passwordhistory

Column	Notes
user_id, password_hash, created_at	Last 5 passwords kept (app logic)

5. Audit, versioning & operations

tuition_auditlog

Column	Notes
timestamp, action, model_name, record_id	CREATE, UPDATE, DELETE, LOGIN, etc.
user_id, user_ip, user_agent	Who / where
field_name, old_value, new_value, description	Change detail
metadata, session_id, request_id	JSON (or text on SQLite)

tuition_dataversion

Column	Notes
model_name, record_id, version_number	Unique together
data_snapshot, created_at, created_by_id	Full record snapshot

tuition_systemhealth

Column	Notes
timestamp, component, status, message, metrics	HEALTHY / WARNING / CRITICAL

tuition_securityevent

Column	Notes
timestamp, event_type, severity, description	Security alerts
user_id, user_ip, user_agent, metadata	
resolved, resolved_at, resolved_by_id	Resolution tracking

6. Django framework tables (also present)

Standard Django apps create additional tables, for example: `django_session`, `django_admin_log`, `django_content_type`, `auth_group`, `auth_permission`, and join tables for user groups/permissions.

PK Primary key **FK** Foreign key **Unique** Unique constraint

Generated from repository schema · Regenerate: [open docs/database-schema.html](#) in Chrome → Print → Save as PDF, or run the project's PDF build script if added.